DeepWatch: A Unified Multimodal Platform for Deepfake Forensic Analysis and Generation-Method Attribution

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Abstract

Deepfake media is increasingly used for fraud and fabricated evidence, yet published research addresses detection, attribution, and speech anti-spoofing as separate problems, and no reviewed system produces explainable, integrity-protected forensic output. This proposal analyses four IEEE/CVF publications, maps each documented limitation to a corresponding platform component, and proposes **DeepWatch**: a forensic pipeline combining detection, open-set generation-method attribution, explainable evidence, and integrity-preserving reporting. All components are proposed methods to be evaluated against the base papers' published benchmarks; no experimental results are claimed at this stage.

Keywords *— Deepfake detection, multimedia forensics, generation-method attribution, audio–video fusion, evidence integrity.*

1. Introduction and Problem Statement

Generative models allow non-experts to fabricate convincing faces, voices, and video. Documented misuse includes voice-clone fraud and manipulated visual material. Existing detection tools output opaque probability scores trained on controlled datasets; they neither explain findings nor produce evidence a victim or investigator can reuse. This proposal therefore targets two questions: whether media shows signs of manipulation, and whether that assessment can be documented in a forensically defensible way.

2. Literature Review and Research Gap

Four IEEE-venue papers form the technical foundation. Each contributes a component capability; each also documents a limitation relevant to this proposal.

Base paper	Approach and contribution	Documented limitation → DeepWatch mapping
UCF, Yan et al., ICCV 2023 [1]	Disentangles forgery-irrelevant, method-specific, and common forgery features; detection uses only common features for cross-generator generalization. Reports 0.996 AUC in-dataset but 0.828 mean AUC across unseen datasets; requires per-method forgery labels.	**L1:** generalization degrades cross-dataset; label cost; no explanation → **D1** detection module targets improved cross-dataset AUC on the same benchmarks; **D4** adds explainable outputs.
Up-sampling artifacts, Tan et al., CVPR 2024 [2]	Shows that CNN generators' up-sampling layers leave detectable artifacts; uses these for generalizable detection.	**L2:** method is tied to CNN-generator artifact classes and does not address generators lacking them (e.g., diffusion architectures) → **D2** adds complementary frequency/noise analysis, evaluated on generator classes outside [2]'s scope.
OW-DFA / CPL, Sun et al., ICCV 2023 [3]	Introduces open-world deepfake attribution benchmark; contrastive pseudo learning attributes fakes to methods unseen in training.	**L3:** benchmarked on facial forgeries only; attribution accuracy on unseen methods remains limited → **D3** extends open-set attribution and reports explicit "unseen method" abstention, evaluated on the OW-DFA protocol.
ASVspoof 2021, Liu et al., IEEE/ACM TASLP 2023 [4]	Standard community benchmark and baselines for spoofed and deepfake **speech** detection.	**L4:** evaluates the audio channel in isolation → **D2** fuses speech analysis with visual analysis and lip–speech consistency for video-call media.

Research gap. Within the reviewed literature, detection [1][2], attribution [3], and speech anti-spoofing [4] are evaluated as separate tasks; the surveyed sources do not present an integrated pipeline that performs all three and emits explainable, integrity-protected evidence. This review is scoped to the cited sources and comparable recent work and does not claim to be exhaustive.

3. Research Objectives and Proposed Methodology

Each proposed component answers a documented limitation. These are planned methods, not verified results.

- **D1 — Detection module.** Common-forgery-feature detection following [1], trained with compression and resolution augmentation; *evaluation:* AUC/EER on FaceForensics++ in-dataset and cross-dataset on Celeb-DF and DFDC, reported against [1]'s published baselines.
- **D2 — Multimodal fusion.** Visual score from D1 combined with a speech anti-spoofing score trained on [4]'s protocol plus a lip–speech synchronisation measure; *evaluation:* fused vs. unimodal AUC/EER on bimodal test media, including compressed variants.
- **D3 — Open-set attribution.** Contrastive pseudo learning after [3], extended with an explicit abstention class for out-of-distribution generators; *evaluation:* top-k attribution accuracy and open-set rejection metrics on the OW-DFA protocol.
- **D4 — Explainable evidence.** Region-level artifact localisation (heatmaps) with plain-language descriptions, presented as indicators supporting assessment — not proof of manipulation or authenticity.
- **D5 — Forensic reporting.** Structured PDF report bundling all scores, visualisations, hashes, and analysis metadata for investigators.

4. Digital Forensic Framework and Evaluation Plan

Evidence integrity. On submission, each file is hashed (SHA-256); hash, timestamp, and provenance are logged before analysis. Original media is preserved unmodified; processing runs on copies. Every analysis step, model version, and parameter is written to an append-only analysis log for traceability and re-examination. Report hashes may optionally be anchored on a public test blockchain as tamper evidence — noting that anchoring attests to a report's integrity after creation, not to the truth of its contents.

Evaluation. Detection is benchmarked against [1] and [2] on their published datasets and splits; attribution against [3] on OW-DFA; the speech channel against [4]'s partitions. Metrics: AUC and EER for detection; top-k accuracy and open-set metrics for attribution; expected calibration error for confidence outputs. False-positive and false-negative rates are reported separately — false accusations and missed forgeries carry different real-world costs and are not collapsed into a single figure. All code, configurations, model hashes, and evaluation scripts will be released for third-party reproduction.

5. Expected Contributions, Challenges, and Conclusion

Planned contributions: (a) an evaluated detect–attribute–explain–report pipeline prototype; (b) an audio–video fusion method measured on compression-degraded media; (c) a reproducible evaluation against published baselines. **Challenges:** cross-dataset generalization remains an open problem, so improvement targets are stated modestly; compression destroys low-level artifacts; detection outputs are probabilistic indicators, and consequential decisions still require human review. **Conclusion:** the four base papers supply validated components whose documented limitations jointly justify this work; DeepWatch is proposed to address them as a unified platform, with all claims subject to experimental verification against published benchmarks.

References

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